

$$5) \ y = -\frac{x^3}{x^2 - 4}$$

$$6) \ y = (5x + 30)^{\frac{2}{3}}$$

$$7) \ y = -\frac{3}{16}(x-1)^{\frac{4}{3}} - \frac{3}{2}(x-1)^{\frac{1}{3}} + 2$$

Critical thinking question:

- 8) Sketch a continuous curve $y = f(x)$ where $f(1) = 0$, $f'(0) = 0$, $f'(2) = 0$, $f''(x) < 0$ for $x < 1$, and $f''(x) > 0$ for $x > 1$.

